2022-23 - First semester
Financial time series (SFA01)
Master 2, principal session

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Exercise 1. Let us consider a time series $\overrightarrow{Y} = ((Y_t(k))_{1\le k\le m})_{t\in\mathbb{Z}}$ valued in $\mathbb{R}^m$, defined by

$$Y(k) = \mathrm{F}_{\psi^{(k)}}(X) + \sigma\,\epsilon(k)\;,\qquad k = 1,\ldots,m\;,$$

where $\sigma > 0$ and

(i) $X = (X_t)_{t\in\mathbb{Z}}$ is a centered time series valued in $\mathbb{R}$ with spectral density function f^X;

(ii) $\psi^{(k)} = \left(\psi_t^{(k)}\right)_{t\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ and $\mathrm{F}_{\psi^{(k)}}$ is the associated convolutional filter defined by

$$\left[\mathrm{F}_{\psi^{(k)}}(X)\right]_t = \sum_{s\in\mathbb{Z}} \psi_s^{(k)} X_{t-s}\;,\qquad t\in\mathbb{Z}\;;$$

(iii) $\overrightarrow{\epsilon} = ((\epsilon_t(k))_{1\le k\le m})_{t\in\mathbb{Z}} \sim \mathrm{WN}(0,\Sigma)$ is a centered white noise process valued in $\mathbb{R}^m$ with covariance matrix Σ, uncorrelated with X. We assume moreover that Σ is positive definite (hence invertible).

1. Give the spectral density matrix function of $\overrightarrow{\epsilon}$.
2. Give, for $k = 1,\ldots,m$, the spectral density function of $\mathrm{F}_{\psi^{(k)}}(X)$ using the Fourier series functions
$$\psi^{(k)*}(\lambda) = \sum_{t\in\mathbb{Z}} \psi_t^{(k)}\,\mathrm{e}^{-\mathrm{i}\lambda t}\;,\qquad k = 1,\ldots,m\;,\ \lambda\in\mathbb{R}.$$
3. Show that the spectral density matrix function M of $\overrightarrow{Y}$ reads
$$M(\lambda) = f^X(\lambda)\,\overrightarrow{a}(\lambda)\overrightarrow{a}(\lambda)^H + A\;,$$
where $\overrightarrow{a}(\lambda)$ is a vector of $\mathbb{C}^m$ and A is a $m\times m$ matrix, and determine $\overrightarrow{a}(\lambda)$ and A.
4. Deduce the coherence function $C_{k,l} = |M_{k,l}|\,/\sqrt{M_{k,k}\,M_{l,l}}$ for all $1\le k<l\le m$.
5. What is the behavior of $C_{k,l}(\lambda)$ as $\sigma\to 0$? Express the result using the frequency sets defined by
$$\Lambda := \left\{\lambda\ :\ f^X(\lambda)\neq 0\right\}$$
and, for $k = 1,\ldots,m$ by
$$\Lambda_k = \left\{\lambda\ :\ \psi^{(k)*}(\lambda)\neq 0\right\}\;,$$

From now on, we assume that $\sigma = 1$ and $f^X(\lambda) = 1/(2\pi)$ for all λ. We denote by B the backshift operator.

6. Give M and $C_{k,l}$ if, moreover, for all $k = 1, \ldots, m$, $\mathrm{F}_{\psi^{(k)}} = \mathrm{B}^{\tau_k}$ for some $\tau_k \in \mathbb{Z}$?

From now on, we moreover assume that, for all $k = 1, \ldots, m$,

$$\psi^{(k)*}(\lambda) = \frac{\Theta_k(\mathrm{e}^{-\mathrm{i}\lambda})}{\Phi_k(\mathrm{e}^{-\mathrm{i}\lambda})} ,$$

where Θ_k and Φ_k are coprime polynomials with real coefficients such that $\Theta_k(0) = \Phi_k(0) = 1$.

7. What can we say about the process $\mathrm{F}_{\psi^{(k)}}(X)$ for $k = 1, \ldots, m$?

8. Show that the process $\overrightarrow{Y}$ is a VARMA process. [**Hint**: use that a weakly stationary vector process $\overrightarrow{Z} = \left(\overrightarrow{Z}_t\right)_{t\in\mathbb{Z}}$ such that $\mathrm{Cov}\,(Z_s, Z_t) = 0$ for $|t-s| > q$ is a VMA(q) process.]

9. Let Φ be the lowest common multiple of $\Phi_1, \ldots, \Phi_m$. Show that, for all $\overrightarrow{x} \in \mathbb{R}^m$, $\Phi(\mathrm{B})\overrightarrow{x}^T\overrightarrow{Y}$ is an MA process.

Exercise 2. Let X and Y be two jointly $\mathcal{N}(0,1)$ random variables and define $\alpha = \mathrm{Cov}\,(X, Y)$. We recall that two variables X and Y are said to be jointly Gaussian if (X, Y) is a Gaussian vector.

10. Show that $\alpha \in [-1, 1]$ and give the mean and the covariance matrix of the $\mathbb{R}^2$-valued vector (X, Y).

11. Show that $\mathrm{Cov}\,\left(Y^2, Y\right) = 0$ and $\mathrm{Var}\,\left(Y^2\right) = 2$.

12. Determine $\hat{X} = \mathrm{proj}\,(X|\,\mathrm{Span}\,(Y))$, and, setting $\epsilon = X - \hat{X}$, show that ϵ and Y are independent. Deduce that $\mathrm{Cov}\,\left(X^2, Y^2\right) = 2\alpha^2$.

13. Deduce that for two centered jointly Gaussian variables Z_1 and Z_2, we have

$$\mathrm{Cov}\,\left(Z_1^2, Z_2^2\right) = 2\left(\mathrm{Cov}\,(Z_1, Z_2)\right)^2 .$$

14. Conclude that, for four centered jointly Gaussian variables $Z_1, \ldots, Z_4$, we have

$$\mathrm{Cov}\,(Z_1Z_3, Z_2Z_4) = \mathrm{Cov}\,(Z_1, Z_2)\ \mathrm{Cov}\,(Z_3, Z_4) + \mathrm{Cov}\,(Z_1, Z_4)\ \mathrm{Cov}\,(Z_3, Z_2) .$$

[**Hint**: use the identity $ab = \frac{1}{4}\left((a+b)^2 - (a-b)^2\right)$.]

Let us now consider a real valued and centered Gaussian process $(Z_t)_{t\in\mathbb{Z}}$ with autocovariance function γ. We assume that it admits a spectral density function f in $L^2(\mathbb{T}, \mathcal{B}(\mathbb{T}))$, which implies that

$$\sum_{k\in\mathbb{Z}} \gamma^2(k) = 2\pi \int_{-\pi}^{\pi} f^2(\lambda)\,\mathrm{d}\lambda < \infty .$$

Let us define

$$\widetilde{\gamma}_n(h) = \frac{1}{n}\sum_{t=1}^{n-|h|} Z_t\, Z_{t+|h|} \text{ if } |h| < n \text{ and } \widetilde{\gamma}_n(h) = 0 \text{ otherwise} , \tag{1}$$

seen as an estimator of $\gamma(h)$ from $Z_1, \ldots, Z_n$.

15. Compute the bias $b(h) = \mathbb{E}[\widetilde{\gamma}_n(h)] - \gamma(h)$ for all $h \in \mathbb{Z}$.

16. Show that, for all $h \in \mathbb{Z}$,

$$\mathrm{Var}\left(\widetilde{\gamma}_n(h)\right) = \frac{1}{n} \sum_{k \in \mathbb{Z}} (1 - (|h| + |k|)/n)_+ \ \left[\gamma^2(k) + \gamma(k+h)\gamma(k-h)\right] ,$$

where $a_+ = \max(a, 0)$.

17. Derive the limit

$$\ell(h) = \lim_{n \to \infty} n \mathbb{E}\left[\left(\widetilde{\gamma}_n(h) - \gamma(h)\right)^2\right] .$$

and express $\ell(h)$ using f.

We now further assume that

$$\sum_{k \in \mathbb{Z}} \gamma(k) < \infty ,$$

and we suppose that we observe $U_t = Z_t + \mu$, $t = 1, \ldots, n$, with $\mu \in \mathbb{R}$ unknown and we denote the usual empirical covariance estimator by

$$\hat{\gamma}_n(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} \left(U_t - \hat{\mu}_n\right)\left(U_{t+|h|} - \hat{\mu}_n\right) \text{ if } |h| < n \text{ and } \hat{\gamma}_n(h) = 0 \text{ otherwise} ,$$

where $\hat{\mu}_n = \frac{1}{n} \sum_{k=1}^n U_k$. We set for any $\tau \geq 0$, $m_n(\tau) = \frac{1}{n} \sum_{k=1+\tau}^{n-\tau} Z_k$.

18. For $h \in \mathbb{Z}$ and $n > 2|h|$, express $\hat{\gamma}_n(h) - \widetilde{\gamma}_n(h)$ using $m_n(0)$ and $m_n(|h|)$.

19. Let $\tau \in \mathbb{N}$, what is the limit of $n \, \mathbb{E}\left[m_n^4(\tau)\right]$ as $n \to \infty$?

20. What are the limits of $n \, \mathbb{E}\left[\left(\hat{\gamma}_n(h) - \widetilde{\gamma}_n(h)\right)^2\right]$ and $n \, \mathbb{E}\left[\left(\hat{\gamma}_n(h) - \gamma(h)\right)^2\right]$ as $n \to \infty$?

Answers

Solution of Exercice 1 1.

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3.

4.

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8.

9.

Solution of Exercice 2 10. By Cauchy-Schwartz inequality, we have $|\alpha| \leq 1$. The mean of (X, Y) is $(0,0)$ and its covariance matrix is $\begin{bmatrix} 1 & \alpha \\ \alpha & 1 \end{bmatrix}$.

11. We have for $Y \sim \mathcal{N}(0,1)$, $\mathbb{E}[Y^3] = \mathbb{E}[Y] = 0$ and $\mathbb{E}[Y^4] = 3$. The formulas follow.

12. We have $\hat{X} = \alpha Y$ since it belongs to $\mathrm{Span}\,(Y)$ and satisfies the orthogonality condition $\mathbb{E}[(X - \hat{X})Y] = \mathrm{Cov}\left(X - \hat{X}, Y\right) = 0$. Since ϵ and Y are jointly Gaussian and uncorrelated, they are independent.

    Using that ϵ and Y are independent and the previous questions, we have

    $$\mathrm{Cov}\left(X^2, Y^2\right) = \mathrm{Cov}\left((\alpha Y)^2 + \epsilon^2 + 2\alpha\epsilon Y, Y^2\right) = 2\alpha^2 + 0 + 2\alpha \mathrm{Cov}\left(\epsilon Y, Y^2\right) \,.$$

    Since ϵ and Y are independent and centered, $\mathrm{Cov}\left(\epsilon Y, Y^2\right) = \mathbb{E}[\epsilon Y^3] - 0 = \mathbb{E}[\epsilon]\mathbb{E}[Y^3] = 0$ and we get the result.

13. We can write $Z_1 = \sigma_1 X$ and $Z_2 = \sigma_2 Y$ with X and Y as previously and $\mathrm{Cov}\,(Z_1, Z_2) = \sigma_1\sigma_2\alpha$. The bilinearity of covariance and the result of the previous question implies

    $$\mathrm{Cov}\left(Z_1^2, Z_2^2\right) = \sigma_1^2\sigma_2^2 \mathrm{Cov}\left(X^2, Y^2\right) = 2(\sigma_1\sigma_2\alpha)^2.$$

    Hence the result.

14. We write Z_1Z_3 and Z_2Z_4 using the hinted formula and get that

$$\begin{aligned}
\mathrm{Cov}(Z_1Z_3, Z_2Z_4) &= \frac{1}{16}\,\mathrm{Cov}\left((Z_1+Z_3)^2-(Z_1-Z_3)^2, (Z_2+Z_4)^2-(Z_2-Z_4)^2\right) \\
&= \frac{1}{16}\left[\mathrm{Cov}\left((Z_1+Z_3)^2, (Z_2+Z_4)^2\right) - \mathrm{Cov}\left((Z_1-Z_3)^2, (Z_2+Z_4)^2\right)\right. \\
&\left. - \mathrm{Cov}\left((Z_1+Z_3)^2, (Z_2-Z_4)^2\right) + \mathrm{Cov}\left((Z_1-Z_3)^2, (Z_2-Z_4)^2\right)\right] \\
&= \frac{1}{8}\left[\mathrm{Cov}^2\left(Z_1+Z_3, Z_2+Z_4\right) - \mathrm{Cov}^2\left(Z_1-Z_3, Z_2+Z_4\right)\right. \\
&\left. - \mathrm{Cov}^2\left(Z_1+Z_3, Z_2-Z_4\right) + \mathrm{Cov}^2\left(Z_1-Z_3, Z_2-Z_4\right)\right] \\
&= \frac{1}{8}\left[\left(\mathrm{Cov}\left(Z_1, Z_2+Z_4\right) + \mathrm{Cov}\left(Z_3, Z_2+Z_4\right)\right)^2\right. \\
&\quad - \left(\mathrm{Cov}\left(Z_1, Z_2+Z_4\right) - \mathrm{Cov}\left(Z_3, Z_2+Z_4\right)\right)^2 \\
&- \left(\mathrm{Cov}\left(Z_1, Z_2-Z_4\right) + \mathrm{Cov}\left(Z_3, Z_2-Z_4\right)\right)^2 \\
&\quad \left. + \left(\mathrm{Cov}\left(Z_1, Z_2-Z_4\right) - \mathrm{Cov}\left(Z_3, Z_2-Z_4\right)\right)^2\right] \\
&= \frac{1}{8}\left[4\,\mathrm{Cov}\left(Z_1, Z_2+Z_4\right)\mathrm{Cov}\left(Z_3, Z_2+Z_4\right)\right. \\
&\left. -4\,\mathrm{Cov}\left(Z_1, Z_2-Z_4\right)\mathrm{Cov}\left(Z_3, Z_2-Z_4\right)\right] \ .
\end{aligned}$$

After developing the covariances with respect to their second arguments and some simplifications, we get the formula that we are looking for.

15. We have, for all $h \in \mathbb{Z}$, $\mathbb{E}[\widetilde{\gamma}_n(h)] = (1-|h|)/n)_+\gamma(h)$ so

$$b(h) = \left[(1-|h|/n)_+ - 1\right]\ \gamma(h) \ .$$

16. Using that Z is a Gaussian process and Question 14, we get, for all $h \in \mathbb{Z}$,

$$\begin{aligned}
\mathrm{Var}\left(\widetilde{\gamma}_n(h)\right) &= \frac{1}{n^2}\sum_{t,t'=1}^{n-|h|} \mathrm{Cov}(Z_tZ_{t+|h|}, Z_{t'}Z_{t'+|h|}) \\
&= \frac{1}{n^2}\sum_{t,t'=1}^{n-|h|} \left(\gamma^2(t-t') + \gamma(t-t'-|h|)\gamma(t-t'+|h|)\right) \\
&= \frac{1}{n}\sum_{k\in\mathbb{Z}}(1-(|h|+|k|)/n)_+\left(\gamma^2(k) + \gamma(k-|h|)\gamma(k+|h|)\right) \ ,
\end{aligned}$$

where we set $k = t - t'$.

17. Applying the previous questions, we have for all $h \in \mathbb{Z}$,

$$\begin{aligned}
n\mathbb{E}\left[(\widetilde{\gamma}_n(h) - \gamma(h))^2\right] &= n\, b^2(h) + n\ \mathrm{Var}\left(\widetilde{\gamma}_n(h)\right) \\
&= n\ \left[(1-|h|/n)_+ - 1\right]^2\ \gamma^2(h) \\
&+ \sum_{k\in\mathbb{Z}}(1-(|h|+|k|)/n)_+\left(\gamma^2(k) + \gamma(k-|h|)\gamma(k+|h|)\right) \ .
\end{aligned}$$

For n large enough, we have $n\ \left[(1-|h|/n)_+ - 1\right]^2 = h^2/n$ which tends to zero as $n \to \infty$. On the other hand, we have

$$\sum_{k\in\mathbb{Z}} \gamma^2(k) < \infty \quad \text{and so} \quad \sum_{k\in\mathbb{Z}} |\gamma(k-|h|)\gamma(k+|h|)| < \infty\ .$$

Hence by dominated convergence, we get

$$\begin{aligned}
\lim_{n\to\infty} \sum_{k\in\mathbb{Z}} &(1-(|h|+|k|)/n)_+ \left(\gamma^2(k) + \gamma(k-|h|)\gamma(k+|h|)\right) \\
&= \sum_{k\in\mathbb{Z}} \lim_{n\to\infty} (1-(|h|+|k|)/n)_+ \left(\gamma^2(k) + \gamma(k-|h|)\gamma(k+|h|)\right) \\
&= \sum_{k\in\mathbb{Z}} \left(\gamma^2(k) + \gamma(k-|h|)\gamma(k+|h|)\right)\ .
\end{aligned}$$

This is our limit $\ell(h)$ of $n\mathbb{E}\left[(\widetilde{\gamma}_n(h) - \gamma(h))^2\right]$ as $n \to \infty$.

By the Parseval and Plancherel identities, we have

$$\sum_{k\in\mathbb{Z}} \gamma^2(k) = 2\pi \int_{-\pi}^{\pi} f^2(\lambda)\, \mathrm{d}\lambda$$

and

$$\sum_{k\in\mathbb{Z}} \gamma(k-|h|)\gamma(k+|h|) = 2\pi \int_{-\pi}^{\pi} f^2(\lambda)\, \mathrm{e}^{2\mathrm{i}|h|\lambda}\ \mathrm{d}\lambda = 2\pi \int_{-\pi}^{\pi} f^2(\lambda)\ \cos(2|h|\lambda)\ \mathrm{d}\lambda$$

(by symmetry of f). Hence we get

$$\ell(h) = 2\pi \int_{-\pi}^{\pi} f^2(\lambda)\, (1+\cos(2|h|\lambda))\ \mathrm{d}\lambda\ .$$

18. Take $n > 2|h|$. Observing that $U_t - \hat{\mu}_n = Z_t - m_n(0)$, we have

$$\begin{aligned}
\hat{\gamma}_n(h) &= \frac{1}{n} \sum_{t=1}^{n-|h|} (Z_t - m_n(0))\,(Z_{t+|h|} - m_n(0)) \\
&= \tilde{\gamma}_n(h) + (1-|h|/n)\ m_n^2(0) - \frac{m_n(0)}{n} \left(\sum_{t=1}^{n-|h|} Z_t + \sum_{t=1}^{n-|h|} Z_{t+|h|} \right)\ .
\end{aligned}$$

It suffices now to observe that for $n > 2|h|$, we have

$$\sum_{t=1}^{n-|h|} Z_t + \sum_{t=1}^{n-|h|} Z_{t+|h|} = \sum_{t=1}^{n-|h|} Z_t + \sum_{t=1+|h|}^{n} Z_t = \sum_{t=1}^{n} Z_t + \sum_{t=1+|h|}^{n-|h|} Z_t\ .$$

We get that

$$\hat{\gamma}_n(h) = \widetilde{\gamma}_n(h) + (1-|h|/n) m_n^2(0) - m_n(0)\ [m_n(0) + m_n(|h|)]\ .$$

19. $m_n(\tau)$ is a centered Gaussian variable with variance

$$n^{-2} \sum_{k,k'=1+\tau}^{n-\tau} \gamma(k-k') \leq n^{-1} \sum_k |\gamma(k)| = O(n^{-1}) .$$

Hence its moment of order 4 is $O(n^{-2})$ as $n \to \infty$.

20. Let $h \in \mathbb{Z}$. From the two previous questions and the Minkowski's inequality, we have that, for $n > 2|h|$,

$$\begin{aligned}\left(\mathbb{E}\left[(\hat{\gamma}_n(h) - \widetilde{\gamma}_n(h))^2\right]\right)^{1/2} &\leq 2\left(\mathbb{E}\left[(m_n(0))^4\right]\right)^{1/2} + \left(\mathbb{E}\left[(m_n(0)m_n(|h|))^2\right]\right)^{1/2} \\ &= O(n^{-1}) ,\end{aligned}$$

where we also used that $\mathbb{E}\left[(m_n(0)m_n(|h|))^2\right] \leq \left(\mathbb{E}\left[(m_n(0))^4\right]\right)^{1/2} \left(\mathbb{E}\left[(m_n(|h|))^4\right]\right)^{1/2}$. We thus get

$$n\,\mathbb{E}\left[(\hat{\gamma}_n(h) - \widetilde{\gamma}_n(h))^2\right] = O(n^{-1}) .$$

Now, using again the Minkowski's inequality, we have

$$\left|\left(n\,\mathbb{E}\left[(\hat{\gamma}_n(h) - \gamma(h))^2\right]\right)^{1/2} - \left(n\,\mathbb{E}\left[(\widetilde{\gamma}_n(h) - \gamma(h))^2\right]\right)^{1/2}\right| \leq \left(n\,\mathbb{E}\left[(\hat{\gamma}_n(h) - \widetilde{\gamma}_n(h))^2\right]\right)^{1/2} = O(n^{-1/2}) .$$

Hence $n\,\mathbb{E}\left[(\hat{\gamma}_n(h) - \gamma(h))^2\right]$ and $\mathbb{E}\left[(\widetilde{\gamma}_n(h) - \gamma(h))^2\right]$ have the same limit as $n \to \infty$, that is, $\ell(h)$.